

Guide: Transition Engine

1 Introduction

The render-transition function is designed to automatically generate “roadmap” or “summary” slides when the document structure changes (e.g., entering a new section).

Previously, it required many manual parameters. Now, it is designed to be **configured globally** via navigator-config, allowing a clean one-line integration in your document.

2 Global Configuration

Instead of passing parameters to every render-transition call, you can set them once at the beginning of your document:

```
#import "@preview/navigator:0.2.0": navigator-config, render-transition
```

```
#navigator-config.update(c => {  
  c.mapping = (section: 1, subsection: 2)  
  c.slide-func = my-presentation-engine.slide  
  c.theme-colors = (primary: navy, accent: orange)  
  c  
})
```

```
// Now you can use a simple one-liner:
```

```
#show heading: render-transition
```

3 Function documentation

render-transition(h, transitions: (:), mapping: (:), ...)

3.1 Parameters Reference

Most parameters default to auto, which means they will be resolved from the global navigator-config state.

Option	Type	Description
h	heading	Mandatory. The heading object intercepted by the show rule.
slide-func	function auto	A callback (fill: color, body: content) => content used to create the slide. If auto, resolved from global config.
transitions	dict	Detailed configuration for the transition engine. These settings are merged with the global config defaults.
mapping	dict auto	Maps heading levels to roles (e.g., (section: 1, subsection: 2)). If auto, resolved from global config.
theme-colors	dict auto	Dictionary containing primary and accent colors. If auto, resolved from global config.
show-heading-numbering	bool auto	Whether to display heading numbers in the roadmap. If auto, resolved from global config.
numbering-format	string auto	Numbering format (e.g., "1.1"). If auto, resolved from global config.
base-text-size	length auto	Base font size for the roadmap text. Default: auto.
base-text-font	string auto	Font family for the roadmap text. Default: auto.

Option	Type	Description
top-padding	relative length	Vertical spacing added above the roadmap in Standard Mode . Default: 40%.
content-padding	length dict	Padding around the roadmap content in Standard Mode . Default: (x: 10%).
content-align	alignment	Alignment of the roadmap content in Standard Mode . Default: top + left.
content-wrapper	function	A callback (roadmap, heading, active-state) => content to completely override the slide layout (Expert Mode).
max-length	int dict auto	Maximum length of titles before truncation. If auto, resolved from global config.
use-short-title	bool dict auto	Whether to use short titles. If auto, resolved from global config.

3.2 The transitions dictionary

This parameter allows fine-tuning the behavior and appearance of transition slides.

Key	Type	Description
enabled	bool	Global switch for transitions. Default: true.
max-level	int	The maximum heading level that triggers a transition. Default: 3.
background	color string	Background type: "theme" (uses primary color), "none", or an explicit color.
filter	function	A callback (heading) => bool to programmatically enable/disable transitions for specific headings.
style	dict	Controls typography: inactive-opacity (default 0.3), completed-opacity (default 0.6), active-weight (default "bold").
sections	dict	Override for section-level transitions. Contains enabled, visibility (dict), and background.
subsections	dict	Override for subsection-level transitions. Contains enabled, visibility (dict), and background.

3.2.1 Visibility Logic

For each transition role (parts, sections, subsections), you can define which hierarchy levels are visible using the `visibility` key. The behavior depends on the active heading:

- "all": Show all headings at this level throughout the document.
- "current": Only show the heading that is currently active (or the parent of the active one).
- "current-parent": Show all siblings of the active heading (i.e., all headings at this level that share the same parent).
- "none": Completely hide this level from the roadmap.

3.3 Customizing the Layout

You have two ways to control the roadmap appearance on the slide:

1. **Standard Mode:** Use `top-padding`, `content-padding`, and `content-align` to position the roadmap. This is quick and works for most cases.
2. **Expert Mode:** Use `content-wrapper` to take full control. Note that if `content-wrapper` is provided, the standard padding and alignment parameters are ignored.

4 Basic usage

4.1 Explicit mode (Legacy/Direct)

You can still pass all parameters manually if you prefer total isolation for specific headings.

```
render-transition(  
  h,  
  mapping: (section: 1, subsection: 2),  
  theme-colors: (primary: navy),  
  slide-func: my-slide-func,  
)
```

Direct Parameter Passing

2 Global Configuration

4.2 Minimalist mode (Recommended)

By using `navigator-config.update(...)`, your show rule becomes extremely clean:

`#show heading: render-transition`

This is the recommended way to use Navigator in modern presentations.

5 Layout & Positioning Control

5.1 Padding and Alignment

Use `content-padding` and `content-align` to place the roadmap exactly where you want it.

```
render-transition(  
  h,  
  top-padding: 0pt,  
  content-align: center + horizon,  
  content-padding: 0pt,  
  slide-func: my-slide-func,  
)
```

Centered Roadmap

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5.2 The Wrapper Mode (Total Control)

The `content-wrapper` parameter offers complete freedom over the slide layout. It allows you to place the roadmap alongside other elements (images, titles, decorations) or to wrap it in complex layouts.

The callback function receives three arguments:

1. `roadmap`: The progressive outline component generated by Navigator.
2. `heading`: The specific heading object that triggered this transition.
3. `active-state`: A dictionary `{ h1, h2, h3 }` containing the active hierarchy (useful to retrieve the parent Part title when entering a Section).

5.2.1 Example 1: Split Layout (Title + Roadmap)

In this example, we place the current section title on the left and the roadmap on the right. We also disable the section display in the roadmap to avoid repetition.

```
content-wrapper: (roadmap, h, active) => {
  grid(
    columns: (1fr, 1.5fr),
    align(center + horizon,
      text(size: 1.5em, weight: "bold",
        format-heading(h))
    ),
    align(left + horizon, roadmap)
  )
},
transitions: (
  sections: (visibility: (section: "none",
    subsection: "all"))
)
```

Split Layout

2 Global Configuration

3.1 Parameters Reference
3.2 The transitions dictionary

5.2.2 Example 2: Contextual Header

Use active-state to display the parent heading (e.g., the Part) above the roadmap.

```
content-wrapper: (roadmap, h, active) => {
  set align(left + top)
  if active.h1 != none {
    text(size: 0.8em, fill:
      white.transparentize(40%),
    smallcaps(active.h1.body))
    v(0.5em)
  }
  roadmap
}
```

Contextual Info

Function documentation

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3.1 Parameters Reference

3.2 The transitions dictionary

5.2.3 Example 3: Branding (Logo + Roadmap)

You can easily integrate graphical elements.

```
content-wrapper: (roadmap, h, active) => {
  stack(dir: ltr, spacing: 2em,
    align(horizon, image('logo.png')),
    align(horizon, roadmap)
  )
}
```

Branding

1 Introduction

6 Advanced Customization

6.1 Selective visibility

You can configure different roadmap layouts for sections and subsections.

```
transitions: (  
  subsections: (  
    visibility: (  
      section: 'current',  
      subsection: 'current-parent'  
    )  
  )  
)
```

Subsection roadmap

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3.1 Parameters Reference
3.2 The `transitions` dictionary

7 Short Titles & Truncation

`render-transition` inherits the short title and truncation capabilities of `progressive-outline`. This is especially useful for roadmap slides where titles can be very long.

```
render-transition(  
  h,  
  use-short-title: true,  
  max-length: 15,  
  slide-func: my-slide-func,  
)
```

Short Titles & Truncation

2 Global Configur...